MSG No. 222.98 $Pn\bar{3}n$ [Type I, cubic]

Table 1: Wyckoff site: 2a, site symmetry: 432

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]

Table 2: Wyckoff site: 6b, site symmetry: 42.2

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,2,3,8,9,10,13,14]
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[4,5,15,16,18,19,21,23]
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[6,7,11,12,17,20,22,24]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[25,26,27,32,33,34,37,38]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[28,29,39,40,42,43,45,47]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[30,31,35,36,41,44,46,48]

Table 3: Wyckoff site: 8c, site symmetry: .-3.

No.	position	mapping
1	$[0, 0, 0]$	[1,17,18,25,41,42]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,7,15,26,31,39]
3	$\left[0, 0, \frac{1}{2}\right]$	[3,4,11,27,28,35]
4	$\left[\frac{1}{2}, 0, 0\right]$	[5,6,13,29,30,37]
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[8,22,23,32,46,47]
6	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[9,19,24,33,43,48]
7	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[10,20,21,34,44,45]
8	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[12,14,16,36,38,40]

Table 4: Wyckoff site: 12d, site symmetry: -4..

No.	position	mapping
1	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[1,8,26,27]
2	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[2,3,25,32]
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[4,16,43,47]
4	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[5,15,42,45]
5	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[6,11,41,48]
6	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[7,12,44,46]
7	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[9,10,37,38]
8	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[13,14,33,34]
9	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[17,24,30,35]
10	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[18,21,29,39]
11	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[19,23,28,40]
12	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[20,22,31,36]

Table 5: Wyckoff site: `12e`, site symmetry: `4..`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,2,3,8]`
2	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	`[4,16,19,23]`
3	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	`[5,15,18,21]`
4	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[6,11,17,24]`
5	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[7,12,20,22]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[9,10,13,14]`
7	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[25,26,27,32]`
8	$\left[\frac{3}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	`[28,40,43,47]`
9	$\left[\frac{3}{4},\ \frac{3}{4},\ -x\right]$	`[29,39,42,45]`
10	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[30,35,41,48]`
11	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[31,36,44,46]`
12	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[33,34,37,38]`

Table 6: Wyckoff site: `16f`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,17,18]`
2	$\left[x,\ \frac{1}{2}-x,\ x\right]$	`[2,7,15]`
3	$\left[x,\ x,\ \frac{1}{2}-x\right]$	`[3,4,11]`
4	$\left[\frac{1}{2}-x,\ x,\ x\right]$	`[5,6,13]`
5	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[8,22,23]`
6	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-x\right]$	`[9,19,24]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$	`[10,20,21]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[12,14,16]`
9	$[-x,\ -x,\ -x]$	`[25,41,42]`
10	$\left[-x,\ x+\frac{1}{2},\ -x\right]$	`[26,31,39]`
11	$\left[-x,\ -x,\ x+\frac{1}{2}\right]$	`[27,28,35]`
12	$\left[x+\frac{1}{2},\ -x,\ -x\right]$	`[29,30,37]`
13	$\left[-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[32,46,47]`
14	$\left[x+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$	`[33,43,48]`
15	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$	`[34,44,45]`
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[36,38,40]`

Table 7: Wyckoff site: `24g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[1,8]`
2	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2,3]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	`[4,16]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ x\right]$	`[5,15]`
5	$\left[\frac{3}{4},\ x,\ \frac{1}{4}\right]$	`[6,11]`
6	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[7,12]`
7	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9,10]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[13,14]`

continued ...

Table 7

No.	position	mapping
9	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[17,24]
10	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[18,21]
11	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[19,23]
12	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[20,22]
13	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[25,32]
14	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[26,27]
15	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[28,40]
16	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[29,39]
17	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[30,35]
18	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[31,36]
19	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[33,34]
20	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[37,38]
21	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[41,48]
22	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[42,45]
23	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[43,47]
24	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[44,46]

Table 8: Wyckoff site: 24h, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{4}, y, y\right]$	[1,13]
2	$\left[\frac{1}{4}, \frac{1}{2}-y, y\right]$	[2,10]
3	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	[3,9]
4	$\left[y, y, \frac{1}{4}\right]$	[4,18]
5	$\left[\frac{1}{2}-y, y, \frac{1}{4}\right]$	[5,19]
6	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	[6,20]
7	$\left[y, \frac{1}{4}, y\right]$	[7,17]
8	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}-y\right]$	[8,14]
9	$\left[y, \frac{1}{4}, \frac{1}{2}-y\right]$	[11,22]
10	$\left[\frac{1}{2}-y, \frac{1}{4}, \frac{1}{2}-y\right]$	[12,24]
11	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	[15,23]
12	$\left[\frac{1}{2}-y, \frac{1}{2}-y, \frac{1}{4}\right]$	[16,21]
13	$\left[\frac{3}{4}, -y, -y\right]$	[25,37]
14	$\left[\frac{3}{4}, y+\frac{1}{2}, -y\right]$	[26,34]
15	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	[27,33]
16	$\left[-y, -y, \frac{3}{4}\right]$	[28,42]
17	$\left[y+\frac{1}{2}, -y, \frac{3}{4}\right]$	[29,43]
18	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	[30,44]
19	$\left[-y, \frac{3}{4}, -y\right]$	[31,41]
20	$\left[\frac{3}{4}, y+\frac{1}{2}, y+\frac{1}{2}\right]$	[32,38]
21	$\left[-y, \frac{3}{4}, y+\frac{1}{2}\right]$	[35,46]
22	$\left[y+\frac{1}{2}, \frac{3}{4}, y+\frac{1}{2}\right]$	[36,48]
23	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	[39,47]
24	$\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{3}{4}\right]$	[40,45]

Table 9: Wyckoff site: 48i, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x,\ \frac{1}{2}-z,\ y\right]$	[2]
3	$\left[x,\ z,\ \frac{1}{2}-y\right]$	[3]
4	$\left[z,\ y,\ \frac{1}{2}-x\right]$	[4]
5	$\left[\frac{1}{2}-z,\ y,\ x\right]$	[5]
6	$\left[\frac{1}{2}-y,\ x,\ z\right]$	[6]
7	$\left[y,\ \frac{1}{2}-x,\ z\right]$	[7]
8	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[8]
9	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[9]
10	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[10]
11	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[12]
13	$\left[\frac{1}{2}-x,\ z,\ y\right]$	[13]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[14]
15	$\left[z,\ \frac{1}{2}-y,\ x\right]$	[15]
16	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	[16]
17	$[z,\ x,\ y]$	[17]
18	$[y,\ z,\ x]$	[18]
19	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[19]
20	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[20]
21	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[21]
22	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[22]
23	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[23]
24	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[24]
25	$[-x,\ -y,\ -z]$	[25]
26	$\left[-x,\ z+\frac{1}{2},\ -y\right]$	[26]
27	$\left[-x,\ -z,\ y+\frac{1}{2}\right]$	[27]
28	$\left[-z,\ -y,\ x+\frac{1}{2}\right]$	[28]
29	$\left[z+\frac{1}{2},\ -y,\ -x\right]$	[29]
30	$\left[y+\frac{1}{2},\ -x,\ -z\right]$	[30]
31	$\left[-y,\ x+\frac{1}{2},\ -z\right]$	[31]
32	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[32]
33	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[33]
34	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[34]
35	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[35]
36	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[36]
37	$\left[x+\frac{1}{2},\ -z,\ -y\right]$	[37]
38	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[38]
39	$\left[-z,\ y+\frac{1}{2},\ -x\right]$	[39]
40	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	[40]
41	$[-z,\ -x,\ -y]$	[41]
42	$[-y,\ -z,\ -x]$	[42]
43	$\left[y+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	[43]
44	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -y\right]$	[44]
45	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	[45]
46	$\left[-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[46]

continued ...

Table 9

No.	position	mapping
47	$\left[-y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[47]
48	$\left[z+\frac{1}{2},\, -x,\, y+\frac{1}{2}\right]$	[48]